



National  
Qualifications  
2014

**X757/75/11**

**Physics  
Relationships Sheet**

THURSDAY, 22 MAY

9:00 AM – 11:00 AM



$$E_p = mgh$$

$$d = vt$$

$$E_k = \frac{1}{2}mv^2$$

$$v = f\lambda$$

$$Q = It$$

$$T = \frac{1}{f}$$

$$V = IR$$

$$A = \frac{N}{t}$$

$$R_T = R_1 + R_2 + \dots$$

$$D = \frac{E}{m}$$

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$$

$$H = Dw_R$$

$$V_2 = \left( \frac{R_2}{R_1 + R_2} \right) V_s$$

$$\dot{H} = \frac{H}{t}$$

$$\frac{V_1}{V_2} = \frac{R_1}{R_2}$$

$$s = vt$$

$$P = \frac{E}{t}$$

$$d = \bar{v}t$$

$$P = IV$$

$$s = \bar{v}t$$

$$P = I^2 R$$

$$a = \frac{v-u}{t}$$

$$P = \frac{V^2}{R}$$

$$W = mg$$

$$E_h = cm\Delta T$$

$$F = ma$$

$$p = \frac{F}{A}$$

$$E_w = Fd$$

$$\frac{pV}{T} = \text{constant}$$

$$E_h = ml$$

$$p_1V_1 = p_2V_2$$

$$\frac{p_1}{T_1} = \frac{p_2}{T_2}$$

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

# Additional Relationships

## Circle

$$\text{circumference} = 2\pi r$$

$$\text{area} = \pi r^2$$

## Sphere

$$\text{area} = 4\pi r^2$$

$$\text{volume} = \frac{4}{3}\pi r^3$$

## Trigonometry

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

